SK015
SET 1

Semester 1
Session 2026/2027
2 hours

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PRE – PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI

CHEMISTRY

SK 015

SET 1

DO NOT OPEN THE QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

RELATIVE ATOMIC MASSES OF SELECTED ELEMENTS
JISIM ATOM RELATIF UNSUR-UNSUR TERPILIH

Element	Symbol	Proton Number	Relative Atomic Mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Fluorine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmium	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	121.8
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.9
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Ionisation constant for water at 25 °C *Hasil darab ion bagi air pada* 25 °C	K_w	=	1.00×10^{-14} mol^2 dm^{-6}
Molar volume of gases *Isi padu molar gas*	V_m	= =	22.4 dm^3 mol^{-1} at STP 24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum *Laju cahaya dalam vakum*	c	=	3.0×10^8 m s^{-1}
Specific heat of water *Muatan haba tentu air*		= = =	4.18 kJ kg^{-1} K^{-1} 4.18 J g^{-1} K^{-1} 4.18 J g^{-1} °C^{-1}
Avogadro's number *Nombor Avogadro*	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant *Pemalar Faraday*	F	=	9.65×10^4 C mol^{-1}
Planck's constant *Pemalar Planck*	h	=	6.6256×10^{-34} J s
Rydberg constant *Pemalar Rydberg*	R_H	= =	1.097×10^7 m^{-1} 2.18×10^{-18} J
Molar of gases constant *Pemalar gas molar*	R	= =	8.314 J K^{-1} mol^{-1} 0.08206 L atm mol^{-1} K^{-1}
Density of water *Ketumpatan air*	ρ	=	1 g cm^{-3}
Freezing point of water *Takat beku air*		=	0.00 °C
Vapour pressure of water *Tekanan wap air*	P_{water}	=	23.8 torr

UNIT AND CONVERSION FACTOR
UNIT DAN FAKTOR PERTUKARAN

Volume *Isi padu*	1 liter = 1 dm^3 1 mL = 1 cm^3
Energy *Tenaga*	1 J = 1 kg m^2 s^{-2} = 1 N m = 10^7 erg 1 calorie = 4.184 Joule 1 eV = 1.602×10^{-19} J
Pressure *Tekanan*	1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101325 N m^{-2}
Others *Lain-lain*	1 Faraday (F) = 96500 Coulomb (C) 1 Newton (N) = 1 kg m s^{-2}

Answer all questions.

1. (a) An ion Q^{2-} has 18 electrons and 16 neutrons.

(i) Determine the proton number and nucleon number of element Q.

(ii) Write the isotopic notation for ion Q^{2-}.

(iii) Element Q has four naturally occurring isotopes. The isotopic masses and percentage abundances are shown in Table 1. Calculate the average atomic mass of element Q.

TABLE 1

Isotope	Isotopic mass (u)	Percentage abundance (%)
^{32}Q	31.972	94.99
^{33}Q	32.971	0.75
^{34}Q	33.968	4.25
^{36}Q	35.967	0.01

[5 marks]

(b) Caffeine is a compound containing carbon, hydrogen, nitrogen and oxygen. The percentage composition by mass of caffeine is 49.48% carbon, 5.19% hydrogen, 28.85% nitrogen and 16.48% oxygen.

(i) Determine the empirical formula of caffeine.

(ii) The molar mass of caffeine is $194\ g\ mol^{-1}$.Determine the molecular formula of caffeine.

(iii) In an experiment, $12.0\ g$ of caffeine is dissolved in water to produce $400\ mL$ solution. The density of the solution is $1.05\ g\ mL^{-1}$. Calculate the molality of the caffeine solution.

[9 marks]

(c) Ammonia gas reacts with oxygen gas to produce nitrogen monoxide and water according to the following equation:

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(l)$$

In an experiment, $6.80\ g$ ammonia is mixed with $12.8\ g$ oxygen.

(i) Determine the limiting reactant.

(ii) Calculate the theoretical mass of nitrogen monoxide produced.

(iii) If $8.16\ g$ of nitrogen monoxide is obtained, calculate the percentage yield of nitrogen monoxide.

[8 marks]

2. (a) Consider electronic transitions within the Lyman series of a hydrogen atom.

(i) Draw an energy level diagram illustrating the electronic transitions that produce the first four lines of this series.

(ii) Determine the transition within this series that has been depicted in a(i), which has the shortest wavelength, and calculate its value.

[6 marks]

(b) Atom X has an atomic number (proton number) of 18.

(i) State the number of electrons with l = 0.

(ii) Sketch the exact shape and orientation of the valence orbital(s) characterised by l = 1.

[4 mark]

3. (a) Beryllium chloride, $BeCl_2$, and sulphur hexafluoride, SF_6, are covalent compounds.

(i) Draw the Lewis structures of $BeCl_2$ and SF_6.

(ii) Explain why both compounds do not obey the octet rule.

(iii) By using orbital diagrams, illustrate the hybridisation of the central atom in SF_6. State the molecular shape. Draw and label the overlapping of orbitals in SF_6.

[12 marks]

(b) Calcium is a metal and a good conductor of electricity.

(i) By using an electron sea model, explain the formation of metallic bonds in calcium.

(ii) Based on band theory, explain the electrical conductivity of calcium.

[5 marks]

4. (a) X_2 and Y_2 react according to the following equation

$$3\,X_2(g) + Y_2(g) \rightarrow 2\,X_3Y\,(g)$$

2 moles of X_2 are allowed to react with 1 mole of Y_2 in a 30 L container at 25 °C. If all X_2 is converted to X_3Y, calculate the total pressure after the reaction is complete.

[6 marks]

b (i) Predict, with reasons, which liquid has a greater surface tension, ethanol (CH_3CH_2OH) or dimethyl ether (CH_3OCH_3).

[3 marks]

5. (a) Consider the following exothermic reaction at equilibrium inside a cylinder fitted with a movable piston:

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

Predict and explain the effect on the equilibrium position and the pressure of SO_3 if argon gas is added to the system at:

(i) Constant volume.

(ii) Constant total pressure.

[4 marks]

(iii) Initially, an unknown amount of SO_3 gas was introduced into a reaction vessel at 700 K. After reaching equilibrium, the mixture contains 0.60 M SO_3, 0.30 M SO_2 and 0.15 M O_2. Calculate the initial pressure of SO_3 gas and its degree of dissociation, α.

[5 marks]

6. (a) Methanoic acid, more commonly known as formic acid, is the simplest carboxylic acid with the chemical formula HCOOH.

(i) Given a 0.150 mol dm^{-3} solution of methanoic acid, HCOOH, at 25°C with a pH of 2.28. Calculate the degree of dissociation of methanoic acid.

[4 marks]

(ii) Predict the pH of salt formed when methanoic acid is added to sodium hydroxide. Write appropriate equations involved.

[5 marks]

(b) A 1.00 L buffer solution contains 0.20 M NH_3 ($K_b = 1.77 \times 10^{-5}$) and 0.15 M NH_4Cl. Calculate the pH of the resulting solution after 0.02 mol of NaOH is added to the buffer solution. Assume the volume remains constant.

[5 marks]